

# Full Stack Development Summary

## Chapter 10

### UTIL Package – Collection Framework

#### What is Util Package

The **Java.util** package is a one of a most important build-in package in a Java.

- ✚ Data storage (Collections)
- ✚ Working with dates
- ✚ Working with random numbers
- ✚ Reading input
- ✚ Handling arrays
- ✚ Using scanners, lists, maps, sets, etc.

#### Why do we need util.package in Java

Because core java does not have build-in features for many task like :

- + Storing multiple object easily
- + Sorting data
- + Getting user input
- + Working with dates and time
- + Generating random numbers

## **Common Classes in java.util Package**

| Class Name  | Purpose                            |
|-------------|------------------------------------|
| Scanner     | To take input from user            |
| ArrayList   | Growable array                     |
| HashMap     | Key-value pair map                 |
| HashSet     | Set of unique values               |
| LinkedList  | List with nodes                    |
| Collections | Utility methods (sort, reverse...) |
| Date        | Work with dates                    |
| Random      | Generate random values             |
| Arrays      | Utility methods for arrays         |

## What is Collections Framework?

Java Collections Framework is a set of **classes and interfaces** that help store, manage, and process groups of objects efficiently.

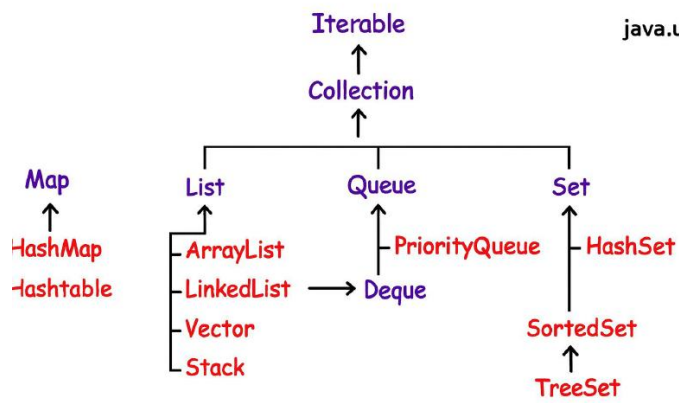
- ✚ Resizable data structures
- ✚ Built-in algorithms (sorting, searching, etc.)
- ✚ Easy data processing
- ✚ Provides ready-made classes like ArrayList, HashSet, HashMap

## Why collections framework

Collections solve the limitations of arrays:

| Arrays                   | Collections                  |
|--------------------------|------------------------------|
| Fixed size               | Grow/shrink dynamically      |
| Can store only same type | Can store any reference type |
| No built-in algorithms   | Has sorting, searching, etc. |
| Manual operations        | Ready-made utility methods   |

## Collection framework



java.util.x



Interfaces that extend Collection:

- List (ordered, duplicates allowed)
- Set (no duplicates)
- Queue (FIFO behavior)

Note: Map is NOT part of Collection.

(Map is a separate interface in the Java Collection Framework.)